

Project Proposal

Multi-view Approach for Detecting Clickbait Headlines

- Abhijit (IMT2023054)
- Vishnu (IMT2023097)
- Adwait (IMT2023502)
- Praneeth (IMT2023555)

1. Introduction

Many of the current models for detecting clickbait are classified as either a clickbait or non-clickbait based on a binary classification. Each headline is labelled as either clickbait or non-clickbait by one model with the only measures of performance being the accuracy and F1 scores of the model used. While these methods can provide reasonable results, they do not provide insight into why a particular headline was labelled as clickbait, nor do they identify which linguistic factors contributed to the model's decision.

In this project, we will expand the scope of clickbait detection from a single binary model to multiple analyses of clickbait headlines from both linguistic and semantic perspectives. Our goal is to not only classify each headline correctly, but also to be able to identify the variety of patterns that contribute to a headline being labelled as clickbait.

2. Project Scope

The project focuses on binary classification of textual headlines into clickbait and non-clickbait categories using an existing labelled dataset. The scope is limited to text-based analysis and does not include images or videos. We will use Pretrained models and standard NLP libraries. The emphasis of the project is on comparison and analysis rather than only improving accuracy.

3. Planned Approach

We are proposing a multi-view approach in which clickbait detection is performed using 3 complementary views:

1. Lexical View:

This view will capture surface-level patterns like TF-IDF features, word n-grams, punctuation usage, headline length and common clickbait phrases. Traditional ML classifiers will be used for this view.

2. Syntactic View:

This view will focus mainly on grammar structure and how the sentence is formed, not the meaning. Simple dependency patterns and parts of speech tags will be extracted by using NLP parsing tools to study stylistic differences between clickbait and non-clickbait headlines.

3. Semantic View:

This view will capture the contextual meaning using pretrained transformer-based sentence embeddings like Sentence BERT or BERT.

Each view will be modelled independently. The predictions from all views will then be combined using an ensemble strategy. We will then compare performance of individual views and the combined system using standard evaluation metrics such as accuracy, precision, recall, F1 score etc.

4. Libraries and Tools

We are planning of using the following libraries and tools to implement this project:

1. Python for implementation
2. Scikit learn for traditional machine learning models and evaluation
3. spaCy for linguistic processing such as part of speech tagging and dependency parsing
4. Hugging Face Transformers for pretrained transformer models
5. Sentence Transformers for semantic sentence embeddings
6. Pandas and NumPy for data processing and analysis

Instead of creating models from scratch, these libraries help us to concentrate on analysis and comparison.

5. Expected Outcomes

The following are the expected outcomes of this project:

1. A working multi-view clickbait detection system.

2. Comparison of lexical, syntactic, and semantic approaches.
3. Understanding which linguistic factors most influence clickbaits.
4. Analysis of limitations and errors of existing approaches.